

TODO List

tests

- ☐ assemble
 - ☐ element
 - ☐ node
- ☒ adjacency
 - ☒ node adjacency
 - ☒ element adjacency

benchmark

- ☐ assemble speed / memory
- ☐ pipeline speed / memory

Example

- ☐ poisson
 - ☒ naive
 - ☐ adaptive mesh
- ☐ heat
 - ☒ naive
 - ☐ pinn
- ☐ wave
 - ☒ naive
 - ☐ pinn
- ☐ linear elasticity
 - ☒ naive
 - ☐ dynamic
- ☐ Fluid mechanics

torch_fem

Function

- ☒ condense
- ☐ quadrature
 - ☒ infinite quadrature for euclidean element
 - ☒ large enough quadrature for triangle
 - ☒ large enough quadrature for tetra
 - ☐ large enough quadrature for wedge
- ☐ shape

- ☐ line
 - ☒ 2 order of shape fn and shape grad
 - ☐ infinite order of shape fn and shape grad
- ☐ triangle
 - ☒ 2 order of shape fn and shape grad
 - ☐ infinite order of shape fn and shape grad
- ☐ quadrilateral
 - ☒ 2 order of shape fn and shape grad
 - ☐ infinite order of shape fn and shape grad
- ☐ tetra
 - ☒ 2 order of shape fn and shape grad
 - ☐ infinite order of shape fn and shape grad
- ☐ brick
 - ☐ 2 order of shape fn and shape grad
 - ☐ infinite order of shape fn and shape grad
- ☐ wedge
 - ☐ 2 order of shape fn and shape grad
 - ☐ infinite order of shape fn and shape grad
- ☐ mesh
 - ☐ adjacency(for gnn)
 - ☐ node adjacency
 - ☐ edge adjacency
 - ☐ mixed mesh
- ☒ assembler
 - ☒ element assembler
 - ☒ node assembler
- ☐ gnn
- ☐ ODE
- ☐ dataset
 - ☐ mesh
 - ☐ generator
 - ☐ gmsh backend
 - ☐ distmesh backend
 - ☒ (hollow)rectangle
 - ☒ (hollow)circle
 - ☒ Lshape
 - ☒ (hollow)cube
 - ☒ (hollow)sphere

- ☐ airfoil/aircraft
- ☐ sparse matrix
 - ☒ spmv/spmm
 - ☒ spsolve
 - ☐ elementwise-op
 - ☒ same layout
 - ☐ different layout
 - ☐ partition
 - ☐ io
 - ☐ det
 - ☐ is_pos_definite
- ☐ strong form to weak form

Efficiency

- ☐ quadrature loop
- ☐ PETsc backend
- ☐ distributed mesh
 - ☐ distributed mesh assemble
 - ☐ distributed linear system solve

Bugs/others

- ☐ retain-graph = True issue fix